

02YMECH - Solution 2

Assigned for the week of Sep 29, 2025

Solutions

1. Uniform circular motion.

A particle moves counter-clockwise on a circle of radius $R = 5$ m with speed $v = 10$ m/s. The center of the circle is at $(x_c, y_c) = (3, 4)$ m. At $t = 0$, the particle is at $(8, 4)$, i.e. 5 units to the right of the center, on the positive x -side.

(a) *Position vector as a function of time.*

Angular speed:

$$\omega = \frac{v}{R} = \frac{10}{5} = 2 \text{ rad/s.}$$

Because the motion is counter-clockwise and starts at angle 0 relative to the $+x$ direction from the center, we can parametrize:

$$x(t) = x_c + R \cos(\omega t) = 3 + 5 \cos(2t),$$

$$y(t) = y_c + R \sin(\omega t) = 4 + 5 \sin(2t).$$

Therefore, the position vector is

$$\mathbf{r}(t) = (3 + 5 \cos(2t)) \hat{i} + (4 + 5 \sin(2t)) \hat{j}.$$

At $t = 0$,

$$\mathbf{r}(0) = (3 + 5, 4 + 0) = (8, 4),$$

which matches the given initial condition.

(b) *Velocity, acceleration at $t = \pi/2$, and perpendicularity.*

Velocity is time derivative:

$$\begin{aligned} \mathbf{v}(t) &= \frac{d\mathbf{r}}{dt} \\ &= \frac{d}{dt} [(3 + 5 \cos(2t)) \hat{i} + (4 + 5 \sin(2t)) \hat{j}] \\ &= (-10 \sin(2t)) \hat{i} + (10 \cos(2t)) \hat{j}. \end{aligned}$$

Its magnitude:

$$\|\mathbf{v}(t)\| = \sqrt{(-10 \sin(2t))^2 + (10 \cos(2t))^2} = \sqrt{100(\sin^2(2t) + \cos^2(2t))} = 10 \text{ m/s,}$$

as expected.

Acceleration is the derivative of velocity:

$$\begin{aligned}\mathbf{a}(t) &= \frac{d\mathbf{v}}{dt} \\ &= \frac{d}{dt} [(-10 \sin(2t)) \hat{i} + (10 \cos(2t)) \hat{j}] \\ &= (-20 \cos(2t)) \hat{i} + (-20 \sin(2t)) \hat{j}.\end{aligned}$$

Evaluate at $t = \pi/2$:

$$2t = 2 \cdot \frac{\pi}{2} = \pi, \quad \sin(\pi) = 0, \quad \cos(\pi) = -1.$$

Then

$$\mathbf{v}\left(\frac{\pi}{2}\right) = (-10 \sin \pi) \hat{i} + (10 \cos \pi) \hat{j} = (0) \hat{i} + (10 \cdot (-1)) \hat{j} = -10 \hat{j} \text{ m/s},$$

$$\mathbf{a}\left(\frac{\pi}{2}\right) = (-20 \cos \pi) \hat{i} + (-20 \sin \pi) \hat{j} = (-20 \cdot (-1)) \hat{i} + (-20 \cdot 0) \hat{j} = 20 \hat{i} \text{ m/s}^2.$$

Now check the dot product at $t = \pi/2$:

$$\mathbf{v}\left(\frac{\pi}{2}\right) \cdot \mathbf{a}\left(\frac{\pi}{2}\right) = (-10 \hat{j}) \cdot (20 \hat{i}) = 0.$$

So they are perpendicular at that instant.

In fact, for *all* t ,

$$\begin{aligned}\mathbf{v}(t) \cdot \mathbf{a}(t) &= (-10 \sin(2t))(-20 \cos(2t)) + (10 \cos(2t))(-20 \sin(2t)) \\ &= 200 \sin(2t) \cos(2t) - 200 \sin(2t) \cos(2t) \\ &= 0.\end{aligned}$$

Thus $\mathbf{v}(t) \perp \mathbf{a}(t)$ for all time. This is characteristic of uniform circular motion: acceleration is purely centripetal (radially inward), velocity is purely tangential.

2. Two particles in the plane.

The position vectors are

$$\mathbf{r}_1(t) = (1 + 2t) \hat{i} + (3 - t) \hat{j}, \quad \mathbf{r}_2(t) = (4 - t) \hat{i} + (1 + 3t) \hat{j}.$$

(a) *Velocities and accelerations.*

Differentiate each with respect to t .

For particle 1:

$$\mathbf{v}_1(t) = \frac{d\mathbf{r}_1}{dt} = 2 \hat{i} + (-1) \hat{j} = 2 \hat{i} - \hat{j},$$

$$\mathbf{a}_1(t) = \frac{d\mathbf{v}_1}{dt} = \mathbf{0}.$$

So particle 1 moves at constant velocity, no acceleration.

For particle 2:

$$\mathbf{v}_2(t) = \frac{d\mathbf{r}_2}{dt} = (-1) \hat{i} + 3 \hat{j} = -\hat{i} + 3 \hat{j},$$

$$\mathbf{a}_2(t) = \frac{d\mathbf{v}_2}{dt} = \mathbf{0}.$$

So particle 2 also moves at constant velocity, no acceleration.

(b) *Time of closest approach and minimum distance.*

The displacement from particle 2 to particle 1 is

$$\Delta \mathbf{r}(t) = \mathbf{r}_1(t) - \mathbf{r}_2(t).$$

Compute components:

$$\Delta x(t) = (1 + 2t) - (4 - t) = 1 + 2t - 4 + t = 3t - 3,$$

$$\Delta y(t) = (3 - t) - (1 + 3t) = 3 - t - 1 - 3t = 2 - 4t.$$

Thus

$$\Delta \mathbf{r}(t) = (3t - 3) \hat{i} + (2 - 4t) \hat{j}.$$

The square of the distance between them is

$$D^2(t) = \|\Delta \mathbf{r}(t)\|^2 = (3t - 3)^2 + (2 - 4t)^2.$$

Expand:

$$(3t - 3)^2 = 9t^2 - 18t + 9,$$

$$(2 - 4t)^2 = 16t^2 - 16t + 4.$$

So

$$D^2(t) = (9t^2 - 18t + 9) + (16t^2 - 16t + 4) = 25t^2 - 34t + 13.$$

To minimize the distance, we minimize $D^2(t)$ (same t since square root is monotonic).

Take derivative:

$$\frac{dD^2}{dt} = 50t - 34.$$

Set to zero:

$$50t - 34 = 0 \quad \Rightarrow \quad t = \frac{34}{50} = \frac{17}{25} = 0.68 \text{ s}.$$

Now evaluate the minimum distance. First compute D^2 at $t = 17/25$:

$$D^2\left(\frac{17}{25}\right) = 25 \left(\frac{17}{25}\right)^2 - 34 \left(\frac{17}{25}\right) + 13.$$

Compute term by term:

$$\left(\frac{17}{25}\right)^2 = \frac{289}{625}, \quad 25 \cdot \frac{289}{625} = \frac{7225}{625} = 11.56.$$

Next:

$$34 \cdot \frac{17}{25} = \frac{578}{25} = 23.12.$$

So

$$D^2\left(\frac{17}{25}\right) = 11.56 - 23.12 + 13 = 1.44.$$

Therefore

$$D_{\min} = \sqrt{1.44} = 1.2 \text{ m}.$$

Summary: the two particles are closest at

$$t = \frac{17}{25} \text{ s} = 0.68 \text{ s}, \quad D_{\min} = 1.2 \text{ m}.$$

Interpretation: since both have constant (but different) velocities, their relative motion is uniform, so the closest approach happens when the line between them is perpendicular to the relative velocity.

3. Explosion and conservation of momentum.

Total mass before breakup:

$$M = 13 \text{ kg}.$$

Masses after breakup:

$$m_1 = 3 \text{ kg}, \quad m_2 = 4 \text{ kg}, \quad m_3 = 6 \text{ kg},$$

with speeds immediately after breakup:

$$\mathbf{v}_1 = (4 \text{ m/s}) \hat{i} \quad (\text{east}), \quad \mathbf{v}_2 = (3 \text{ m/s}) \hat{j} \quad (\text{north}).$$

Before breakup, the body was at rest, so initial total momentum was zero:

$$\mathbf{p}_{\text{total, initial}} = \mathbf{0}.$$

Momentum is conserved, so the vector sum of the fragment momenta after breakup must still be zero:

$$m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 + m_3 \mathbf{v}_3 = \mathbf{0}.$$

Solve for $\mathbf{v}_3$:

$$m_3 \mathbf{v}_3 = -(m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2).$$

First compute the known momenta:

$$m_1 \mathbf{v}_1 = (3 \text{ kg})(4 \text{ m/s}) \hat{i} = 12 \hat{i} \text{ kg} \cdot \text{m/s},$$

$$m_2 \mathbf{v}_2 = (4 \text{ kg})(3 \text{ m/s}) \hat{j} = 12 \hat{j} \text{ kg} \cdot \text{m/s}.$$

Thus

$$m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 = 12 \hat{i} + 12 \hat{j} \quad (\text{kg} \cdot \text{m/s}).$$

So

$$m_3 \mathbf{v}_3 = -(12 \hat{i} + 12 \hat{j}) = (-12) \hat{i} + (-12) \hat{j}.$$

Since $m_3 = 6 \text{ kg}$,

$$\mathbf{v}_3 = \frac{1}{6}(-12 \hat{i} - 12 \hat{j}) = (-2) \hat{i} + (-2) \hat{j} \quad \text{m/s}.$$

So the third fragment moves with velocity

$$\mathbf{v}_3 = (-2, -2) \text{ m/s}.$$

Speed:

$$v_3 = \sqrt{(-2)^2 + (-2)^2} = \sqrt{4 + 4} = \sqrt{8} = 2\sqrt{2} \approx 2.83 \text{ m/s}.$$

Direction: both components are negative, so it moves toward the southwest, at 45° below the $-x$ axis (i.e. equally south and west).

We can express the direction angle θ measured from due west toward south:

$$\tan \theta = \frac{2}{2} = 1 \quad \Rightarrow \quad \theta = 45^\circ.$$

Thus m_3 moves to the southwest at $2\sqrt{2} \text{ m/s}$.

4. **Block on horizontal surface with time-dependent force.**

Given:

$$m = 4 \text{ kg}, \quad \mu_s = 0.5, \quad \mu_k = 0.3, \quad g = 10 \text{ m/s}^2.$$

Applied horizontal force:

$$F(t) = 2t \text{ N}.$$

Normal force:

$$N = mg = (4)(10) = 40 \text{ N}.$$

Maximum static friction magnitude:

$$f_s^{\max} = \mu_s N = 0.5 \times 40 = 20 \text{ N}.$$

Kinetic friction magnitude (once sliding):

$$f_k = \mu_k N = 0.3 \times 40 = 12 \text{ N}.$$

(a) *When does the block start to move?*

While the block is at rest, friction adjusts to match the applied force $F(t)$ up to a maximum of 20 N. The block will start moving when $F(t)$ exceeds $f_s^{\max}$.

We solve

$$F(t) = 20 \text{ N} \quad \Rightarrow \quad 2t = 20 \quad \Rightarrow \quad t = 10 \text{ s}.$$

Interpretation: for $0 \leq t < 10$ s, static friction is enough to hold it in place, so velocity is zero. At $t = 10$ s, motion begins.

(b) *Acceleration immediately after it starts moving.*

Just after $t = 10$ s, the block is in motion, so kinetic friction applies with constant magnitude $f_k = 12$ N opposing the motion.

The applied force at $t = 10$ s is

$$F(10) = 2(10) = 20 \text{ N}.$$

Net horizontal force just after motion begins:

$$F_{\text{net}} = F - f_k = 20 - 12 = 8 \text{ N}.$$

Newton's second law:

$$a = \frac{F_{\text{net}}}{m} = \frac{8}{4} = 2 \text{ m/s}^2.$$

So the block's initial acceleration (once it breaks free) is 2 m/s^2 .

(c) *Velocity 5 seconds after it starts moving.*

After $t = 10$ s, the block continues to be pushed by $F(t) = 2t$. For $t > 10$ s, it is sliding, so friction is kinetic and constant at 12 N.

For $t > 10$ s:

$$F_{\text{net}}(t) = F(t) - f_k = 2t - 12.$$

Then the acceleration for $t > 10$ is

$$a(t) = \frac{F_{\text{net}}(t)}{m} = \frac{2t - 12}{4} = \frac{1}{2}t - 3 \quad (\text{m/s}^2).$$

Integrate from 10 to 15 (the velocity at $t = 10$ is 0):

$$\begin{aligned}v(15) - v(10) &= \int_{10}^{15} \left(\frac{1}{2}t - 3 \right) dt \\&= \left[\frac{1}{4}t^2 - 3t \right]_{10}^{15} \\&= \frac{65}{4} \text{ m/s.}\end{aligned}$$